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INFORMATION-THEORETIC ANALYSIS OF KNOWLEDGE
DISTILLATION

BY

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SENIOR THESIS

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ABSTRACT

Knowledge Distillation (KD) has emerged as a cornerstone technique for transferring the learned representations of high-capacity "teacher" models to more efficient "student" architectures. While the empirical efficacy of KD is well-documented, a rigorous theoretical characterization of its underlying principles remains nascent. This thesis adopts an information-theoretic framework to investigate the fundamental limits of knowledge distillation, with a particular eye towards the compression of Large Language Model (LLM) architectures. By leveraging Rate-Distortion theory, we characterize the information-theoretic optimality of KD as a compression mechanism and establish achievability results for simple model classes. We further supplement this with analysis on performance limits of student models when trained via KD by leveraging techniques from learning theory. Finally, we introduce and formalize the concept of "antidistillation"—the strategic prevention of knowledge transfer—and provide both a theoretical formulation and practical methodologies for its implementation.

To my parents and friends, for their love and support.

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LIST OF ABBREVIATIONS

KD	Knowledge Distillation
LLM	Large Language Model
ML	Machine Learning
R-D	Rate-Distortion
MI	Mutual Information
SOTA	State-Of-The-Art

CHAPTER 1

INTRODUCTION

1.1 Motivation: Efficiency vs Accuracy

Modern Machine Learning (ML) is currently defined by an era of "scaling laws," where improvements in model performance are closely tied to increases in model parameters, computational resources, and training data ([1]). This paradigm has led to state-of-the-art (SOTA) performance across a diverse range of tasks, particularly with the emergence of Large Language Models (LLMs) that exhibit remarkable generalization capabilities. However, the utility of these models is often bounded by the physical realities of their deployment. Their immense scale entails prohibitive memory requirements, high inference latency, and substantial energy consumption, creating a stark conflict between predictive power and practical viability.

To bridge this gap, significant research has focused on developing efficient model implementations that maintain SOTA performance within restricted resource budgets. These efforts generally fall under the umbrella of model compression. As categorized by [2], these methods include pruning and low-rank factorization (which reduce parameter count), quantization (which reduces numerical precision), and knowledge distillation (a transfer-learning approach) ([3]).

Among these, Knowledge Distillation (KD) has empirically emerged as a singularly effective technique for compressing large-scale models. Yet, a theoretical gap remains: while methods like quantization and factorization benefit from relatively mature mathematical frameworks (see [4] and [5]), the success of KD remains largely elusive and poorly understood from a first-principles perspective. This thesis addresses this gap by deriving fundamental information-theoretic limits for Knowledge Distillation. By characterizing these boundaries, we aim to transition KD from a purely empirically

driven field to a principled framework grounded in the rigor of information theory.

1.2 Knowledge Distillation: An Overview

Knowledge Distillation (KD) [3] is a technique where a highly-performant teacher model is used to train a more compact and efficient student model. Unlike the classical supervised learning setting where a model learns from “hard” labels (e.g. true prediction value or classification), in KD, the student model learns from the “soft” outputs of the teacher model. This may be through using the actual outputs of the teacher model as the ground-truth labels for training or by using the “soft” labels from the neuron values in the teacher model immediately before the final output.

While there are many empirical results showing the impressive capability of knowledge distillation for downstream efficient learning, the theoretical understanding of why it works is lacking ([6]). This work provides a unique analysis of knowledge distillation as a compression algorithm and as a specific form of empirical risk minimization to arrive at its information-theoretic limits.

1.3 Contributions and Thesis Organization

This thesis provides a principled framework for understanding Knowledge Distillation through the lens of Information Theory. Our primary contributions include the derivation of rate-distortion and information-theoretic generalization bounds for KD and the formalization of “antidistillation.” The remainder of this work is organized as follows:

Chapter 2 establishes the necessary foundations in Information Theory and Rate-Distortion (R-D) Theory. We review the history of information-theoretic applications in ML, specifically the Information-Bottleneck principle, PAC-Bayes theory, and information-theoretic generalization bounds, before detailing the mechanics of modern model compression.

Chapter 3 presents the core theoretical results. We first derive an R-D framework for KD from a compression perspective, characterizing the R-D

curves for simple models. We then pivot to a learning-theoretic perspective, deriving information-theoretic generalization bounds to bound student performance.

Chapter 4 introduces the concept of “Antidistillation”—a method for preventing a student model from being able to learn from a teacher model. We provide a formulation of this complementary problem and present preliminary results on a practical method for its implementation.

Finally, Chapter 5 summarizes our findings, discusses their place in the broader ML landscape, and outlines future avenues for research.

CHAPTER 2

BACKGROUND AND RELATED WORK

2.1 Information Theory and Rate-Distortion Theory Fundamentals

2.1.1 Information Theory Fundamentals

This section provides an incredibly brief outline of important definitions and results in Information Theory, with a particular emphasis on Rate-Distortion Theory. Much of this exposition follows [7]—refer to them for a more in-depth analysis of information theory and specifically its applications to modern statistical problems. For the full generality required for some of the later results on information-theoretic generalization error bounds, all definitions are provided in their full, measure-theoretic generality.

As a starting point for all of the information-theoretic quantities of interest, we first define the *Kullback-Leibler (KL) divergence* between two probability distributions. This will then be used to define many of the subsequent quantities.

Definition 2.1 (Kullback-Leibler (KL) Divergence). *Let P, Q be distributions on $\mathcal{A}$. The KL divergence (or relative entropy) between P and Q is*

$$D(P\|Q) = \begin{cases} \mathbb{E}_Q \left[\frac{dP}{dQ} \log \frac{dP}{dQ} \right] & P \ll Q \\ +\infty & \text{else} \end{cases}$$

Remark 2.2. *We will adopt the convention that $0 \log 0 = 0$*

Now that we have defined KL-divergence between two distributions we are prepared to define Shannon and differential entropy of random variables (RVs) and the mutual information between two RVs.

Definition 2.3 (Shannon Entropy). Consider a **discrete** random variable X . The Shannon entropy

$$\begin{aligned} H(X) &\triangleq \mathbb{E}_{P_X} \left[\log \frac{1}{P_X(X)} \right] \\ &= \sum_{x \in \mathcal{X}} P_X(x) \log \frac{1}{P_X(x)} \end{aligned}$$

In order to extend this to the continuous case, we define differential entropy.

Definition 2.4 (Differential Entropy). The differential entropy of a random vector X is

$$h(X) = h(P_X) \triangleq -D(P_X \| \text{Leb}), \quad (2.1)$$

where Leb is the Lebesgue measure. In particular, if X has probability density function (pdf) p , then $h(X) = \mathbb{E} \left[\log \frac{1}{p(X)} \right]$; otherwise $h(X) = -\infty$. The conditional differential entropy is $h(X|Y) \triangleq \mathbb{E} \left[\log \frac{1}{p_{X|Y}(X|Y)} \right]$ where $p_{X|Y}$ is a conditional pdf.

Finally, we define the mutual information between two RVs.

Definition 2.5 (Mutual Information). Given two random variables X, Y , their mutual information is

$$I(X; Y) \triangleq D(P_{XY} \| P_X P_Y)$$

A definition that is particularly useful for the analysis of communication channels and for the analysis of generalization of learning algorithms is the Markov kernel.

Definition 2.6 (Markov Kernel (or Channel)). A Markov kernel (or Channel) between two measurable spaces $(\mathcal{X}, \mathcal{A})$ and $(\mathcal{Y}, \mathcal{B})$ is a function $P_{Y|X} : \mathcal{B} \times \mathcal{X} \rightarrow [0, 1]$, often written $P_{Y|X} : (\mathcal{X}, \mathcal{A}) \rightarrow (\mathcal{Y}, \mathcal{B})$ (or when the σ -algebras are clear just $P_{Y|X} : \mathcal{X} \rightarrow \mathcal{Y}$), such that

1. For any $x \in \mathcal{X}$, $P_{Y|X}(\cdot|x)$ is a probability measure on $(\mathcal{Y}, \mathcal{B})$.
2. For any measurable $B \in \mathcal{B}$, $x \mapsto P_{Y|X}(B|x)$ is $\mathcal{A}$ -measurable

What follows are a few important theorems important for the results derived in the later chapters of this thesis, before finally giving a brief introduction to Rate-Distortion Theory.

Theorem 2.7 (Gaussian’s maximize entropy (given Cov)). *Let X be a random vector and Σ be its covariance matrix. Without loss of generality, assume X is centered (zero-mean). Then the distribution P_X which maximizes $h(X)$ is the Gaussian distribution. Specifically,*

$$\max_{P_X: \Sigma_X = \Sigma} h(X) = h(\mathcal{N}(0, \Sigma)) = \frac{1}{2} \log((2\pi e)^d \det \Sigma)$$

Theorem 2.8 (Data Processing Inequality). *Consider three random variables X, Y, Z which form the Markov Chain $X - Y - Z$. Then*

$$I(X; Y) \geq I(X; Z)$$

Theorem 2.9 (Fano’s Inequality). *Let $|\mathcal{X}| = M < \infty$ and $X - Y - \hat{X}$ be a Markov Chain. Let $P_e = \mathbb{P}[X \neq \hat{X}]$, then*

$$H(X|Y) \leq P_e \log(M - 1) + H(P_e)$$

where $H(P_e)$ denotes the entropy of the Bernoulli(P_e) random variable.

We now state a useful variational definition of the KL-divergence.

Theorem 2.10 (Donsker-Varadhan Variational Form of KL-divergence). *Let P, Q be probability measures on $\mathcal{X}$ and denote the class of functions $C_Q := \{f : \mathcal{X} \rightarrow \mathbb{R} \cup \{-\infty\} : 0 < \mathbb{E}_Q[\exp(f(X))] < \infty\}$. Then*

$$D(P||Q) = \sup_{f \in C_Q} \mathbb{E}_P[f(X)] - \log \mathbb{E}_Q[\exp(f(X))]$$

2.1.2 Rate-Distortion Theory Fundamentals

The general structure of Rate-Distortion (R-D) Theory, or more fundamentally, lossy compression, is as follows.

Consider an encoder/decoder pair (Enc, Dec) that induces the following Markov Chain

$$X \xrightarrow{\text{Enc}} W \xrightarrow{\text{Dec}} \hat{X}$$

where $X \in \mathcal{X}$ is called the “source”, $W = \text{Enc}(X)$ is the compressed discrete data, and $\hat{X} = \text{Dec}(W) \in \hat{\mathcal{X}}$ is the reconstruction where $\hat{\mathcal{X}}$ need not be the same as $\mathcal{X}$.

For lossy compression, one also needs a function to measure the distortion between the reconstructed RV and the source RV. This is called the distortion measure and is denoted $d : \mathcal{X} \times \hat{\mathcal{X}} \rightarrow \mathbb{R} \cup \{+\infty\}$. For R-D theory, one wants to minimize the expected distortion $\mathbb{E}[d(X, \hat{X})]$.

For classical R-D theory, one considers the vector $X^n = (X_1, \dots, X_n) \in \mathcal{X}^n$ and its reconstruction $\hat{X}^n = (\hat{X}_1, \dots, \hat{X}_n) \in \hat{\mathcal{X}}^n$. We now define the distortion on these sequences to be

$$d(x^n, \hat{x}^n) \triangleq \frac{1}{n} \sum_{i=1}^n d(x_i, \hat{x}_i).$$

Definition 2.11 ($(2^{nR}, n)$ codes). *A $(2^{nR}, n)$ -rate distortion code consists of an encoding function*

$$f_n : \mathcal{X}^n \rightarrow \{1, 2, \dots, 2^{nR}\}$$

and a decoding function

$$g_n : \{1, 2, \dots, 2^{nR}\} \rightarrow \hat{\mathcal{X}}^n.$$

The distortion associated with this $(2^{nR}, n)$ code is defined as

$$D \triangleq \mathbb{E}[d(X^n, g_n(f_n(X^n)))] = \mathbb{E}[d(X^n, \hat{X}^n)]$$

Definition 2.12 (Achievability). *A rate-distortion pair (R, D) is said to be achievable if there exists a sequence of $(2^{nR}, n)$ -rate distortion codes (f_n, g_n) with*

$$\lim_{n \rightarrow \infty} \mathbb{E}[d(X^n, g_n(f_n(X^n)))] \leq D$$

Definition 2.13 (Rate distortion function). *The rate distortion function $R(D)$ is defined to be the infimum of rates R such that (R, D) is in the closure of the achievable rate distortion pairs (R, D) for a given D .*

Definition 2.14 (Information Rate Distortion Function). *The information rate distortion function $R^I(D)$ for a source X with distortion d is defined as*

$$R^I(D) \triangleq \min_{P_{\hat{X}|X} : \mathbb{E}[d(X, \hat{X})] \leq D} I(X; \hat{X})$$

The central theorem of rate distortion theory is the following theorem,

which relates the rate distortion function and the information rate distortion function.

Theorem 2.15 (Operational Interpretation of $R(D)$). *Consider an independent and identically distributed (iid) source X with distribution P_X and bounded distortion function $d : \mathcal{X} \times \hat{\mathcal{X}} \rightarrow \mathbb{R} \cup \{+\infty\}$. Then*

$$R(D) = R^I(D) = \min_{P_{\hat{X}|X} : \mathbb{E}[d(X, \hat{X})] \leq D} I(X; \hat{X})$$

We will see that for our later discussion on compression and knowledge distillation, this theorem will be turned into a definition for further analysis (2.16).

2.2 Information Theory for Machine Learning

2.2.1 Information-Bottleneck Method

The Information-Bottleneck method ([8]) is a framework originally developed to determine information from a source, particularly relevant for a specific task. One of the main benefits of the information-bottleneck method is that it is distortion-agnostic: one does not need to specify a distortion measure as they would need to in classical R-D theory. This makes it particularly useful for lossy compression in cases where the distortion function is non-obvious, such as speech compression. Furthermore, this principle has been applied to analyze the behavior and generalization capabilities of deep neural networks.

The Information Bottleneck method is derived similarly to the Rate-Distortion function $R(D)$. As in R-D theory, we seek a compressed version $\hat{X}$ of X , but in contrast to R-D theory, here we do not measure the performance of the compressed variable through a distortion function but rather through its information content about some downstream task or side information. Specifically, consider a RV Y which is meaningful for whatever downstream performance we desire from the compressed random variable. The Information-Bottleneck method comes from minimizing the Lagrangian

$$\mathcal{L}(p(\hat{x}|x)) = I(\hat{X}; X) - \beta I(\hat{X}; Y). \quad (2.2)$$

Intuitively, we want to minimize the mutual information between the source variable and the compressed variable to achieve maximal compression while maximizing the mutual information between the compressed variable and the variable representative of downstream performance. [8] were able to find a minimizer for 2.2 as well as a set of equations which can be alternatively optimized to converge towards the minimizer.

Of particular interest for this thesis is the further work on using the Information-Bottleneck method to analyze deep neural networks. This work began with [9] and was followed up on in [10]. The main goal of using the Information-Bottleneck method to analyze deep neural networks is to analyze the mutual information between input and layers in the neural network and using this mutual information to either quantify generalization error or describe the behavior of the neural network.

2.2.2 Information-theoretic Generalization Bounds

One of the central studies of machine learning is the minimization of the *generalization error*. This problem concerns bounding the performance gap between a model trained on a finite dataset and one trained on the full data distribution.

Following [11], let $\mathcal{Z}$ be the instance space (e.g. in supervised learning $\mathcal{Z} = \mathcal{X} \times \mathcal{Y}$ where $\mathcal{X}$ and $\mathcal{Y}$ are the feature space and label space respectively). The learning algorithm has an iid training set $\mathbf{Z} = (Z_1, \dots, Z_n) \in \mathcal{Z}^n \sim P_{\mathcal{Z}}^{\otimes n}$. The learner selects a hypothesis $W \in \mathcal{W}$ where $\mathcal{W}$ is the hypothesis space. The method for choosing the hypothesis from the training set is denoted $P_{W|\mathbf{Z}}$. The performance of a specific hypothesis is quantified by a loss function $\ell : \mathcal{W} \times \mathcal{Z} \rightarrow \mathbb{R}^+$. We finally consider the generalization error

$$\Delta(w, \mathbf{Z}) \triangleq \mathbb{E}_{P_{\mathcal{Z}}}[\ell(w, Z)] - \frac{1}{n} \sum_{i=1}^n \ell(w, Z_i). \quad (2.3)$$

The intuition behind studying $\Delta(w, \mathbf{Z})$ is that if one minimizes the empirical loss

$$\frac{1}{n} \sum_{i=1}^n \ell(w, Z_i),$$

then if $\Delta(w, \mathbf{Z})$ is small this minimizer will also perform well on the data

distribution P_Z as a whole.

The most studied form of information-theoretic generalization error bounds in bounding $\mathbb{E}_{P_{WZ}}[\Delta(W, \mathbf{Z})]$, however, in classical PAC (Probably Approximately Correct)-Bayesian literature, one typically bounds $\mathbb{E}_{P_{W|Z}}[\Delta(W, \mathbf{Z})]$ with high probability by imposing a prior π on $\mathcal{W}$. The most practically important—but most complex—form of generalization error bounds is high probability bounds on $\Delta(W, \mathbf{Z})$ with respect to P_{WZ} . This is the most commonly used procedure in practice but imposes massive complexities in terms of actual theoretical analysis.

2.3 Model Compression and Knowledge Distillation

The rapid advancement of deep neural networks (DNNs) has been characterized by a significant increase in parameter count, often leading to models that are computationally prohibitive for deployment on resource-constrained edge devices. Model compression aims to alleviate these constraints by reducing the storage and computational footprint of a model while minimizing the degradation in task performance. Formally, given a large "teacher" model f_T with parameter set θ_T , we seek a compressed "student" model f_S with $|\theta_S| \ll |\theta_T|$ such that the generalization error $\Delta(f_S, \mathbf{Z})$ remains bounded.

2.3.1 Knowledge Distillation

Knowledge Distillation (KD), popularized by [3], is a compression paradigm where a small student model is trained to mimic the behavior of a pre-trained, larger teacher model. Unlike standard supervised learning, which utilizes "hard" one-hot labels, KD leverages the "soft" probabilities produced by the teacher's softmax layer. These soft targets, intuitively, provide a "dark knowledge" richer than hard labels, as they capture the inter-class similarities learned by the teacher. The student is typically trained using a weighted objective function:

$$\mathcal{L}_{KD}(x) = (1 - \alpha)\mathcal{L}_{CE}(y, \sigma(f_S(x))) + \alpha\mathcal{L}_{KL}(\sigma(f_T(x)), \sigma(f_S(x))) \quad (2.4)$$

where $\alpha \in [0, 1]$, $\mathcal{L}_{CE}$ is the cross-entropy loss, $\mathcal{L}_{KL}$ is the KL divergence.

2.3.2 Parameter Pruning and Quantization

While KD focuses on functional mimicry, pruning and quantization focus on structural reduction.

- **Weight Pruning:** This involves removing redundant parameters from the network that contribute minimally to the output. Pruning can be unstructured (removing individual weights, leading to sparse matrices) or structured (removing entire neurons or filters to maintain hardware-friendly dense computations).
- **Quantization:** This technique reduces the precision of the model’s weights and activations. By moving from 32-bit floating-point (FP32) representations to lower bit-widths, such as INT8 or even binary values, one significantly reduces memory bandwidth and power consumption.

A major work on model compression through specifically pruning and quantization is [12]. This work has largely motivated the perspective of this work on an information-theoretic perspective of specifically knowledge distillation. [12] frames model compression within the framework of R-D theory, originally developed for data compression, and uses this framework to derive information-theoretic compression bounds for simple linear and one-hidden-layer ReLU networks. Specifically, they assume there is some distribution P_W of the parameters, and there are infinitely many models whose parameters are i.i.d. from P_W which will be compressed. This is analogous to the R-D interpretation of compression and is not exactly what happens in practice, but it does offer very interesting interpretations and intuitions for model compression.

From this assumption, we then take as a definition, Thm. 2.15 in the setting of model compression. Specifically,

Definition 2.16. *The rate-distortion function for model compression is defined as,*

$$R(D) := \min_{P_{\hat{W}|W}: \mathbb{E}[d(W, \hat{W})] \leq D} I(W; \hat{W})$$

where W is the original model and $\hat{W}$ is the compressed representation.

CHAPTER 3

THE INFORMATION THEORY FOR KNOWLEDGE DISTILLATION

3.1 A Compression Perspective

Consider the knowledge distillation perspective, where we have a “teacher” model which in this paper will be parameterized by $\theta_T \in \mathcal{T} \subseteq \mathbb{R}^t$, which teaches a student model $\theta_S \in \mathcal{S} \subseteq \mathbb{R}^s$. For knowledge distillation, the first goal is for the student model to mirror the behavior of the teacher model. This can be captured by a distortion function $d : \mathcal{S} \times \mathcal{T} \rightarrow \mathbb{R}^+$ which measures the distortion between the teacher and student models. For the use of information-theoretic measures, we will assume that θ_T and θ_S are random variables. The validity of this assumption can be argued through the stochasticity of gradient descent, random initialization, dropout, and many other methods (see [11] for more details on these arguments).

As a secondary goal, one wants the student model to be “smaller” than the teacher model. This notion of smaller can take many forms, however, borrowing from [12], we take the information-theoretic compression perspective and view this notion of “smaller” as minimizing the mutual information between the teacher and student (further discussion on the interpretation of the mutual information term is given in section 3.1.2). Armed with this, we can now present our information-theoretic formulation of knowledge distillation.

Definition 3.1 ((Information-theoretic) Knowledge Distillation). *Given a stochastic “teacher” model $\theta_T \in \mathcal{T} \subseteq \mathbb{R}^t$, a stochastic “student” model $\theta_S \in \mathcal{S} \subseteq \mathbb{R}^s$, and a distortion function $d : \mathcal{S} \times \mathcal{T} \rightarrow \mathbb{R}^+$, we define*

$$D(R) := \min_{P_{\theta_S|\theta_T} : I(\theta_T;\theta_S) \leq R} \mathbb{E}[d(\theta_S, \theta_T)]$$

The notation in Definition 3.1 should be incredibly familiar, as it is very analogous to Definition 2.16 used to define an information-theoretic notion

of model compression. The notation used to denote optimization problem is purposely suggesting: from classical R-D theory, we know that the distortion-rate function $D(R) = R^{-1}(D)$. Thus, since Definition 3.1 is exactly the distortion-rate function, we can see that the solution to the knowledge distillation problem is exactly the inverse of the model compression problem. Thus, from this view, we can see how the two fields are incredibly linked.

Because $D(R) = R^{-1}(D)$, since classical information theory is more focused on $R(D)$, all of our following results will focus on classifying $R(D)$, but can be trivially mapped to $D(R)$.

3.1.1 Derivation of KD Loss

We will now show how Definition 3.1 can directly give rise to the classical knowledge distillation loss function given in equation 2.4.

Theorem 3.2 (Informal; Equivalence of KD definitions). *Under the settings of classical knowledge distillation, the information-theoretic definition above collapses to the classical objective.*

Proof. The proof involves considering the degenerate Dirac measures centered at the given teacher model θ_T^* and at the potential student models $\hat{\theta}_S$. More details are provided in Appendix A. $\square$

Remark 3.3. *One can extend this to the loss function incorporating the cross-entropy term involving the labels y by adequately modifying the distortion function d used.*

3.1.2 Interpretations of Mutual Information Regularization

The mutual information constraint $I(\theta_T; \theta_S) \leq R$ may seem odd at first as it has no direct connection to the deterministic KD case, however, below are some interpretations of this mutual information constraint.

Firstly, if the rate R was infinite, the easiest way for the student to achieve zero distortion (perfectly matching the teacher) would be to simply copy the teacher’s weights exactly. However, this is just memorization. By constraining $I(\theta_T; \theta_S)$, you are “telling” the student to learn a compression/representation of the teacher that still performs well. This, ideally,

forces the student to learn the underlying function or logic that the teacher represents, rather than blindly copying the teacher’s specific parameter configuration. This thus acts as a strict mathematical barrier against overfitting to the teacher’s architecture.

In a similar vein, the mutual information constraint enforces a sort of “Occam’s Razor”—where the student tries to learn the most efficient and simplest way to represent the teacher model. This simpler model ideally generalizes better on unseen data as it is less overfit to the specifics of the samples given to it.

3.2 Characterization of Rate Distortion

3.2.1 R-D Curves for Simple Models

To begin, we present some results on simple classes of models, specifically, linear and logistic models.

Theorem 3.4 (Linear Model KD). *Let $\theta_T \in \mathbb{R}^d$ be the teacher parameter vector drawn from*

$$\theta_T \sim \mathcal{N}(0, \Sigma_T),$$

where Σ_T is positive semi-definite. Let $x \in \mathbb{R}^d$ be the input data vector, drawn from a distribution with zero mean and a strictly positive definite covariance matrix $\Sigma_x \succ 0$. We assume x is statistically independent of θ_T and any student parameter θ_S . Let

$$d(\theta_T, \theta_S) = \mathbb{E}_x \left[(\theta_T^T x - \theta_S^T x)^2 \right]$$

Finally, let $\Sigma_x^{1/2}$ be the unique symmetric positive definite square root of Σ_x . Define the data-weighted teacher covariance matrix as:

$$\tilde{\Sigma}_T = \Sigma_x^{1/2} \Sigma_T \Sigma_x^{1/2} \tag{3.1}$$

Let $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d \geq 0$ be the eigenvalues of $\tilde{\Sigma}_T$.

The minimal expected distortion $D(R)$ is given parametrically by the re-

verse water-filling equations:

$$R(\mu) = \sum_{i=1}^d \frac{1}{2} \max \left(0, \ln \left(\frac{\lambda_i}{\mu} \right) \right) \quad (3.2)$$

$$D(\mu) = \sum_{i=1}^d \min(\mu, \lambda_i) \quad (3.3)$$

where the water level parameter $\mu > 0$ is chosen such that $R(\mu) = R$.

Proof. To prove this, we first transform the problem into a Gaussian rate distortion problem and then apply the classical reverse water-filling technique. Refer to Appendix A for more details. $\square$

Lemma 3.5 (Optimal Channel). *The distortion-rate function $D(R)$ is achieved by the following transition channel $P_{\hat{z}|z_T}$, applied independently to each component in the rotated eigenbasis $z = V^\top \tilde{\theta}$:*

$$\hat{z}_i = \begin{cases} 0 & \text{if } \lambda_i \leq \mu, \\ \left(1 - \frac{\mu}{\lambda_i}\right) z_{T,i} + w_i & \text{if } \lambda_i > \mu, \end{cases} \quad (3.4)$$

where $w_i \sim \mathcal{N}(0, \mu(1 - \mu/\lambda_i))$ is independent of $z_{T,i}$, and $\mu > 0$ is the water level corresponding to rate R . The student weights are recovered via the inverse transformations:

$$\theta_S = \Sigma_x^{-1/2} V \hat{z}. \quad (3.5)$$

Furthermore, this channel is the unique minimizer of the rate-distortion problem.

Proof. To prove this, we simply check that the rate and distortion of this channel match the bounds given in Theorem 3.4 and apply known results on Gaussian channels to arrive at uniqueness. Refer to Appendix A for more details. $\square$

The above two results classify the rate-distortion and distortion-rate function for the problem of distilling linear models via a mean-squared error distortion. This is the canonical formulation as it very cleanly aligns with classical results in Gaussian rate-distortion theory. We extend this result to the logistic regression case below under some additional regularity assumptions.

To begin with, consider the logistic model $f(x; \theta) = \sigma(\theta^T x)$ where $\sigma(\cdot)$ is the sigmoid function. In order to find the rate-distortion function for this paradigm, we will make the following assumptions.

Assumption 3.6 (Prior). *The teacher model parameters $\theta_T \in \mathbb{R}^k$ are drawn from a zero-mean multivariate Gaussian prior: $\theta \sim \mathcal{N}(0, \Sigma_T)$.*

Assumption 3.7 (Distortion Measure). *The distortion between the teacher model θ_T and the student model θ_S is measured by the expected Kullback-Leibler (KL) divergence over the data distribution $p(x)$:*

$$d(\theta, \theta') = \mathbb{E}_{x \sim p(x)} \left[D_{KL} \left(\sigma(\theta^T x) \parallel \sigma(\theta'^T x) \right) \right]$$

where $\sigma(\cdot)$ is the sigmoid function.

Assumption 3.8 (Local Quadratic Approximation). *The student model θ_S is assumed to be sufficiently close to the teacher θ_T such that the KL divergence can be accurately approximated by its second-order Taylor expansion.*

Assumption 3.9 (Constant Fisher Information). *The expected Fisher Information Matrix (FIM), defined as $H = \mathbb{E}_x[\sigma(\theta^T x)(1 - \sigma(\theta^T x))xx^T]$, is assumed to be a constant positive-definite matrix over the local region of approximation.*

Theorem 3.10 (Logistic Model KD). *Under assumptions 3.6–3.9, the minimum expected distortion $D(R)$ achievable by a student model θ_S given a capacity constraint on the mutual information $I(\theta_T; \theta_S) \leq R$ is given by the following parametric equations (the reverse water-filling solution):*

$$R(\mu) = \frac{1}{2} \sum_{i=1}^k \max \left(0, \log \frac{\gamma_i}{\mu} \right)$$

$$D(\mu) = \frac{1}{2} \sum_{i=1}^k \min(\gamma_i, \mu)$$

where $\mu > 0$ is the water-level parameter acting as the Lagrange multiplier, and $\gamma_1, \dots, \gamma_k$ are the eigenvalues of the variance-weighted Fisher Information Matrix $M = \Sigma_\theta^{1/2} H \Sigma_\theta^{1/2}$.

Proof. Using Assumption 3.8 and 3.9 use a Taylor expansion to translate the problem into a Gaussian rate-distortion problem. Then apply the reverse water-filling technique. Refer to Appendix A for more details. $\square$

The above result thus gives the information-theoretic limit of knowledge distillation for simple logistic models. Although these models are not nearly as complex as the modern architectures used today, they can help provide intuition as to the limits of these modern architectures. They also help to provide a clear direction forward towards the analysis of modern (e.g. Transformer) architectures.

3.2.2 Optimality of Knowledge Distillation

From Theorem 3.2, we know that the knowledge distillation loss can be derived from the information-theoretic definition of knowledge distillation, which itself is simply the distortion-rate function. This thus suggests that knowledge distillation may actually be the information-theoretically optimal compression procedure for parametric models (e.g. neural networks). This is because Theorem 3.2 suggests that minimizing the knowledge distillation loss is approximating the rate-distortion function, which is the optimal compression given an allowable distortion level.

This observation may help provide a reason behind the incredible success of knowledge distillation in practice, specifically with LLMs. By distilling knowledge from a teacher model into a smaller student model, you are learning an information-theoretically optimal representation of the weights for the downstream task that you are attempting to perform.

3.3 A Learning Theory Perspective

The preceding section established that knowledge distillation is information-theoretically optimal as a compression procedure: minimizing the KD loss approximates the rate-distortion function, which achieves the minimum possible distortion at a given rate R . However, this optimality is expressed entirely in terms of the *empirical* distortion on the transfer set S . A natural and critical question follows: does this compression optimality extend to unseen data? In other words, does a student that optimally mimics the teacher on S also faithfully represent the teacher’s behavior on the broader data distribution μ ? To answer this, we turn to the information-theoretic generalization theory of [11], which will allow us to connect the rate R from

Definition 3.1 directly to the student’s generalization gap.

3.3.1 Setup and Population Distortion

Let $S = \{x_1, \dots, x_n\}$ be a transfer dataset of n samples drawn i.i.d. from a distribution μ over the input space $\mathcal{X}$. We now define the *empirical distortion* on the transfer set as

$$\hat{d}(\theta_S, \theta_T, S) := \frac{1}{n} \sum_{i=1}^n \ell(\theta_S(x_i), \theta_T(x_i)), \quad (3.6)$$

and the *population distortion* as

$$d_{\text{pop}}(\theta_S, \theta_T) := \mathbb{E}_{X \sim \mu}[\ell(\theta_S(X), \theta_T(X))], \quad (3.7)$$

where $\ell : \mathcal{Y} \times \mathcal{Y} \rightarrow \mathbb{R}^+$ is a per-sample loss function (e.g., mean-squared error or KL divergence) measuring the pointwise disagreement between the student and teacher predictions. The optimization in Definition 3.1 targets d_{pop} ; in practice, however, one only has access to $\hat{d}$. Understanding the gap between these two quantities is therefore essential to understanding the practical utility of the framework.

3.3.2 An Information-Theoretic Generalization Bound for KD

We now apply the bound of [11] to the distillation setting. Abstractly, their result states that if a learning algorithm produces a hypothesis W from a dataset S and the per-sample loss is σ -subgaussian under the data distribution, then

$$|\mathbb{E}[L_{\text{pop}}(W) - L_{\text{emp}}(W, S)]| \leq \sqrt{\frac{2\sigma^2}{n} I(W; S)}. \quad (3.8)$$

We identify $W \leftrightarrow \theta_S$ and $S \leftrightarrow S$ with per-sample loss $\ell(\theta_S(x_i), \theta_T(x_i))$. The subgaussianity condition is standard; it is satisfied whenever ℓ is bounded on $[0, b]$ (with $\sigma = b/2$), which holds for common choices such as bounded cross-entropy or squared-error loss with compact parameter spaces. Applying (3.8)

directly yields

$$\left| \mathbb{E} \left[d_{\text{pop}}(\theta_S, \theta_T) - \hat{d}(\theta_S, \theta_T, S) \right] \right| \leq \sqrt{\frac{2\sigma^2}{n} I(\theta_S; S)}. \quad (3.9)$$

Equation (3.9) bounds the distillation generalization gap directly in terms of how much information the student’s parameters retain about the specific transfer set used to train it.

3.3.3 Connecting to the Rate via the Data Processing Inequality

The bound in (3.9) involves $I(\theta_S; S)$, which is not directly controlled by Definition 3.1. The key step is to connect this quantity to the teacher–student mutual information $I(\theta_T; \theta_S)$, which is the object our rate-distortion framework operates on.

Assumption 3.11 (Distillation Markov Chain). *The student model observes only the teacher’s output representations and not the raw input data directly, so that the following Markov chain holds:*

$$S \rightarrow \theta_T \rightarrow \theta_S. \quad (3.10)$$

That is, conditioned on θ_T , the student parameters θ_S are conditionally independent of S .

Remark 3.12. *Assumption 3.11 is satisfied in response-only distillation protocols, where the student receives only the teacher’s output vectors $\{\theta_T(x_i)\}_{i=1}^n$ without access to the corresponding inputs $\{x_i\}$ (e.g., distillation from anonymized soft labels or from a black-box API). In the more common setting where the student is trained on labeled pairs $\tilde{S} = \{(x_i, \theta_T(x_i))\}$, the chain (3.10) does not hold exactly, since θ_S depends on both the teacher outputs and the raw inputs. In this case, the bound below applies with $I(\theta_S; S)$ replaced by $I(\theta_S; \tilde{S})$, and Assumption 3.11 can be viewed as an approximation that becomes increasingly valid when the teacher bottlenecks nearly all information about S that is useful for the downstream distillation task.*

Under Assumption 3.11, the Data Processing Inequality (DPI) gives

$$I(\theta_S; S) \leq I(\theta_S; \theta_T) = I(\theta_T; \theta_S), \quad (3.11)$$

where the equality follows from the symmetry of mutual information. Substituting (3.11) into (3.9) yields our main generalization result for the KD setting.

Theorem 3.13 (Information-Theoretic Generalization Bound for KD). *Under the σ -subgaussianity condition on ℓ and Assumption 3.11, the distillation generalization gap satisfies*

$$\left| \mathbb{E} \left[d_{\text{pop}}(\theta_S, \theta_T) - \hat{d}(\theta_S, \theta_T, S) \right] \right| \leq \sqrt{\frac{2\sigma^2}{n} I(\theta_T; \theta_S)}. \quad (3.12)$$

Proof. This follows directly from [11] and the DPI. $\square$

3.3.4 The Rate as a Generalization Regulator

Theorem 3.13 already reveals a meaningful connection between the rate-distortion framework and generalization in the KD setting. That is, the rate constraint $I(\theta_T; \theta_S) \leq R$ in Definition 3.1 directly caps the right-hand side of (3.12).

Corollary 3.14 (Rate-Controlled Generalization). *Under the conditions of Theorem 3.13, any student achieving a compression rate $I(\theta_T; \theta_S) \leq R$ satisfies*

$$\left| \mathbb{E} \left[d_{\text{pop}}(\theta_S, \theta_T) - \hat{d}(\theta_S, \theta_T, S) \right] \right| \leq \sqrt{\frac{2\sigma^2 R}{n}}. \quad (3.13)$$

Corollary 3.14 shows that the rate R plays a dual role in Definition 3.1: it simultaneously controls the achievable *distortion* $D(R)$ (via the rate-distortion tradeoff of the previous section) and the *generalization gap* of the student (via (3.13)).

3.3.5 Bounding the Student’s True Task Risk

So far, the framework has measured distortion relative to the teacher—but the ultimate goal is for the student to perform well on the *true task* with

labels Y . Let ℓ_{task} be the task loss and define the student's true task risk as $L(\theta_S) := \mathbb{E}_{(X,Y)}[\ell_{\text{task}}(\theta_S(X), Y)]$, and similarly $L(\theta_T)$ for the teacher.

Proposition 3.15 (Student Task Risk Decomposition). *Suppose the task loss ℓ_{task} satisfies the triangle inequality (e.g., any metric loss, or KL divergence under the Pinsker-type bound). Then*

$$L(\theta_S) \leq \hat{d}(\theta_S, \theta_T, S) + \sqrt{\frac{2\sigma^2 R}{n}} + L(\theta_T), \quad (3.14)$$

where $I(\theta_T; \theta_S) \leq R$.

Proof. By the triangle inequality, $L(\theta_S) \leq d_{\text{pop}}(\theta_S, \theta_T) + L(\theta_T)$. Decomposing the population distortion as $d_{\text{pop}} = (d_{\text{pop}} - \hat{d}) + \hat{d}$ and applying Corollary 3.14 to bound the generalization gap yields the result. $\square$

Proposition 3.15 gives a three-term decomposition of the student's true risk: (i) the empirical distortion on the transfer set, which measures how well the student currently mimics the teacher on observed data; (ii) a compression-controlled generalization gap, determined solely by the rate R and the transfer set size n ; and (iii) the teacher's own irreducible task error $L(\theta_T)$, which sets a fundamental performance floor for any student trained by distillation alone. Observe that this floor cannot be reduced by increasing n or adjusting R —it can only be addressed through improving the teacher or augmenting distillation with direct task supervision.

CHAPTER 4

ANTIDISTILLATION

4.1 Introduction of Antidistillation

The evolution of reasoning in Large Language Models (LLMs) has yielded significant performance improvements on complex, multi-step tasks such as advanced mathematics, software debugging, and strategic planning. By exposing the intermediate reasoning traces, these models not only achieve higher accuracy but also offer a degree of interpretability and trustworthiness. However, providing full reasoning traces poses a security challenge: while knowledge distillation via reasoning traces was proposed to improve training efficiency ([13, 14]), distilled models have since been shown to match and even surpass their teachers at a fraction of the cost ([15]), raising significant intellectual privacy concerns for frontier labs competing at the edge of learnability [16]. These concerns are now reaching the forefront of United States national policy, with recent legislative proposals explicitly targeting the extraction of capabilities from closed-source AI models [17, 18] that essentially circumvent export controls on compute. Furthermore, from a safety perspective, recent work has shown that distilled models often lose the safety alignment trained into the teacher model ([19, 20]). This creates a direct pathway from publicly exposed reasoning traces to unmoderated, potentially dangerous models. As such, this gives rise to the dual problem of this thesis: how can we prevent a student from learning well from a teacher?

Methods have been proposed to address this problem ([21, 22, 23, 22]). While these defensive techniques represent a significant step forward, they are currently bottlenecked by three primary constraints: computational overhead from retraining either the teacher or training an additional model used in the system; a noticeable degradation in the teacher model’s final output performance; and unacceptable level of incoherence in the sampled reasoning

trace. In addition to the practical limitations, the theoretical foundations of distillation defense methods remain significantly underexplored as current approaches largely lack any formal guarantees necessary for reliable deployment.

More importantly, these empirical defenses lack the formal mathematical guarantees necessary for reliable deployment. To address this gap, this chapter moves beyond heuristic mitigation and formally defines the theoretical limits of preventing knowledge transfer.

4.2 From Rate-Distortion to Game Theory

In the preceding chapters, we established that Knowledge Distillation can be framed as a Rate-Distortion problem. In that framework, a student seeks to minimize its distortion relative to the teacher, constrained by a limited capacity or Rate $I(\theta_T; \theta_S)$. Antidistillation is the adversarial dual of this problem. The defender (the teacher) seeks to intentionally modify its output distribution such that any student, restricted by a bounded capacity, is forced to incur an unacceptably high generalization error. However, translating this theoretical limit into a practical defense requires addressing a crucial operational asymmetry: the defender does not know the exact architecture or capacity of the attacking student a priori. Furthermore, the defender is strictly constrained by detectability: the perturbations applied to the data cannot significantly deviate from the true distribution; otherwise, the utility for legitimate users is destroyed. To utilize our information-theoretic limits under these constraints, we cast the antidistillation problem into a game-theoretic framework. By reformulating the interaction as a bi-level robust Stackelberg game, we can begin to derive robust, principled defenses against distillation attacks.

4.3 A Game-Theoretic Formulation of Antidistillation

4.3.1 Notation

D denotes the clean reference dataset; $\tilde{D}$ denotes a candidate poisoned dataset drawn from an admissible perturbation set $\mathcal{B}$ around D ; θ_T and θ_S denote the teacher and student parameters with corresponding parameter spaces Θ_T and Θ_S ; and $\tilde{\mathcal{L}}$ and $\mathcal{L}$ denote the training loss on $\tilde{D}$ and evaluation loss on D , respectively.

4.3.2 Formulation

In prior literature, antidistillation techniques have primarily focused on minimizing the utility of distilling from reasoning traces while preserving teacher performance. However, any practical defense must also account for detectability: if an attacker can identify that a defense is in place, they can adapt their sampling strategy to evade it. In this section, we formalize the antidistillation problem and show how our general setting naturally handles both concerns.

We motivate our framework by first analyzing the related problem of data poisoning, in which an adversary injects corrupted examples into a training set such that any model trained on it performs poorly at test time. The adversary is constrained to perturbations that remain close to legitimate data, so that the attack is not trivially detectable. This interaction can be viewed as a the following Stackelberg game. First, the attacker commits to a poisoned dataset $\tilde{D}$, after which the defender attempts to find the optimal model parameters from it. Observe that the best action for the defender in this game is:

$$\theta^*(\tilde{D}) := \arg \min_{\theta \in \Theta} \tilde{\mathcal{L}}(\theta; \tilde{D}),$$

while the optimal action for the attacker to take is:

$$\tilde{D}^* := \arg \max_{\tilde{D} \in \mathcal{B}} \mathcal{L}(\theta^*(\tilde{D}); D).$$

Therefore, the game can be viewed as the following bi-level optimization

problem:

$$\tilde{D} = \arg \max_{\tilde{D} \in \mathcal{B}} \mathcal{L}(\arg \min_{\theta} \tilde{\mathcal{L}}(\theta; \tilde{D}); D), \quad (4.1)$$

where $\tilde{D}$ is the poisoned dataset, $\mathcal{B}$ is the set of admissible poisoned datasets, D is the dataset for final evaluation, $\mathcal{L}$ is the loss function to evaluate the optimal model θ , and $\tilde{\mathcal{L}}$ is the loss function for evaluating the model on the poisoned dataset. The solution, $(\tilde{D}^*, \theta^*(\tilde{D}^*))$, to (4.1) constitutes the Stackelberg equilibrium for this data poisoning game.

We find that antidistillation fits this template closely: a model provider releases outputs (the “dataset”) that a third party uses to train a student model, and the provider would like those outputs to be useful to legitimate users while being maximally unhelpful as training data for the student. The bi-level structure carries over directly.

That said, antidistillation differs from classical data poisoning in two important ways. First, the architecture of the downstream model to be poisoned is not necessarily a priori known. Previous methods typically assume access to a proxy model ([21, 23]), however such an assumption may fail to hold. As such, instead of just maximizing the harm caused to the optimal model trained on the poisoned data, the goal of antidistillation is to maximize the minimum error across all potential architectures by optimizing the poison in the training data. Second, as a conceptual difference in this setting, the roles of the attacker and defender are flipped from the classical data poisoning formulation: the defender is poisoning the data whereas the attacker is training the optimal model from the poisoned data.

This gives rise to a slightly different Stackelberg game, where the defender is concerned with maximizing the loss over all allowed architectures rather than maximizing the loss over a specific known architecture. This is because the attacker has a priori chosen a specific architecture to finetune; however, the defender does not know that choice of architecture. Formally, the defender seeks a robust poisoned dataset $\tilde{D}_T^*$ such that

$$\tilde{D}_T^* = \arg \max_{\tilde{D}_T \in \mathcal{B}(D; \theta_T, \Theta_T)} \inf_{\mathcal{H} \in \mathcal{F}} \mathbb{E}_{x \sim p_{\text{data}}} \left[\mathcal{L}(h_{\mathcal{H}}^*(\tilde{D}_T); x) \right],$$

where $\mathcal{B}(D; \theta_T, \Theta_T)^1$ is the set of admissible poisoned datasets which can depend on the original dataset produced by the teacher D , on the teacher’s

¹For convenience, this set will be denoted as $\mathcal{B}$ for the remainder of this paper.

parameters θ_T , and on the architecture of the teacher Θ_T . Additionally, $\mathcal{F}$ is the set of admissible function classes (e.g. model architectures), p_{data} is the true data distribution, and $\mathcal{L}$ is the population loss function. Here, $h_{\mathcal{H}}^*(\tilde{D}_T)$ is defined as

$$h_{\mathcal{H}}^*(\tilde{D}_T) := \arg \inf_{h \in \mathcal{H}} \tilde{\mathcal{L}}(h; \tilde{D}_T).$$

In contrast to this robust formulation for the defender, the attacker's objective is simply

$$h_{\mathcal{H}_s}^*(\tilde{D}_T) = \arg \inf_{h \in \mathcal{H}_s} \tilde{\mathcal{L}}(h; \tilde{D}_T),$$

where the attacker wants to minimize, for their specific architecture $\mathcal{H}_s$, the loss on the poisoned dataset.

This formulation provides the defender with a strict theoretical guarantee: as long as the attacker's true architecture satisfies $\mathcal{H}_s \in \mathcal{F}$, the actual population loss accrued, $\mathbb{E}_{x \sim p_{\text{data}}} \left[\mathcal{L} \left(h_{\mathcal{H}_s}^* \left(\tilde{D}_T \right); x \right) \right]$, is bounded from below by the solution to the robust max-inf objective in (4.2).

The above game gives rise to the following robust optimization formulation of antidistillation (from the defender's perspective):

$$\tilde{D}_T = \arg \max_{\tilde{D}_T \in \mathcal{B}} \inf_{\mathcal{H} \in \mathcal{F}} \mathbb{E}_{x \sim p_{\text{data}}} \left[\mathcal{L} \left(h_{\mathcal{H}}^* \left(\tilde{D}_T \right); x \right) \right]. \quad (4.2)$$

Remark 4.1. *Observe that (4.2) fundamentally differs from formulating the problem as directly evaluating the empirical risk minimizer over the entire function space:*

$$\tilde{D}_T = \arg \max_{\tilde{D}_T \in \mathcal{B}} \mathbb{E}_{x \sim p_{\text{data}}} \left[\mathcal{L} \left(\arg \inf_{h \in \mathcal{F}} \tilde{\mathcal{L}}(h; \tilde{D}_T); x \right) \right].$$

If $\mathcal{F}$ is sufficiently expressive, directly evaluating $\arg \inf_{h \in \mathcal{F}}$ allows the model to perfectly memorize the poisoned dataset $\tilde{D}_T$ while performing arbitrarily poorly on the true distribution p_{data} . This makes the defender's task trivial. By placing the $\inf_{\mathcal{H} \in \mathcal{F}}$ outside the population loss, we impose a stricter, more realistic burden on the defender: it must find a universal poison that maximizes the generalization error against the most resilient admissible architecture, rather than merely exploiting a hypothetical overfit model.

Remark 4.2. *In practice, most constraint sets $\mathcal{B}(D; \theta_T, \Theta_T)$ will take the*

form

$$\mathcal{B}_\epsilon(D; \theta_T) := \left\{ \tilde{D}_T : d(\tilde{D}_T(\tilde{\theta}_T), D(\theta_T)) \leq \epsilon, \tilde{\theta}_T \in \Theta_T \right\},$$

where $\theta_T \in \Theta_T$, d is some distortion measure between the perturbed dataset $\tilde{D}_T$, which is perhaps a function of some perturbed teacher model $\tilde{\theta}_T$, and D , which is a function of θ_T . This distortion measure allows one to capture both the difference in performance of the perturbed dataset (or perturbed teacher model) and the detectability of the proposed antidistillation method.

4.3.3 Instantiations of the Antidistillation Game

Within this subsection, we go through consequences of two specific instantiations of the antidistillation game with respect to $\mathcal{F}$.

Student Architecture is Known

We first consider the simplest case where $\mathcal{F} = \{\mathcal{H}\}$. In this case the antidistillation problem (4.2) simplifies to

$$\tilde{D}_T = \arg \max_{\tilde{D} \in \mathcal{B}} \mathbb{E}_{x \sim p_{\text{data}}} \left[\mathcal{L} \left(\arg \inf_{h \in \mathcal{H}} \tilde{\mathcal{L}}(h; \tilde{D}_T); x \right) \right]. \quad (4.3)$$

The antidistillation problem in this case reduces to the data poisoning problem, described above. When the student model architecture is a priori known, literature from data poisoning can be used to adequately determine the “best” poisoning policy.

Defender has Prior over Architectures

Often, defenders will have some prior $P_{\mathcal{F}}$, which describes the likelihood different model architectures are for distillation attacks. For example, a defender may assign a very high probability to Transformer-based architectures, thus encoding useful information to make their defense stronger. Concretely, the space of allowable student architectures may be finite, thus allowing the defender to construct the following probability space $(\mathcal{F}, 2^{\mathcal{F}}, P_{\mathcal{F}})$. We note that in practice, one would have to imbue the space of all admissible student architectures with a σ -algebra, which may be non-trivial.

Regardless, this permits the following relaxation of (4.2):

$$\tilde{D}_T = \arg \max_{\tilde{D}_T \in \mathcal{B}} \mathbb{E}_{\mathcal{H} \sim P_{\mathcal{F}}} \left[\mathbb{E}_{x \sim p_{\text{data}}} \left[\mathcal{L} \left(h_{\mathcal{H}}^* \left(\tilde{D}_T \right); x \right) \right] \right] \quad (4.4)$$

$$= \arg \max_{\tilde{D}_T \in \mathcal{B}} \mathbb{E}_{(\mathcal{H}, x) \sim P_{\mathcal{F}} \otimes p_{\text{data}}} \left[\mathcal{L} \left(h_{\mathcal{H}}^* \left(\tilde{D}_T \right); x \right) \right]. \quad (4.5)$$

² Also, observe that

$$\begin{aligned} & \max_{\tilde{D}_T \in \mathcal{B}} \inf_{\mathcal{H} \in \mathcal{F}} \mathbb{E}_{x \sim p_{\text{data}}} \left[\mathcal{L} \left(h_{\mathcal{H}}^* \left(\tilde{D}_T \right); x \right) \right] \\ & \leq \max_{\tilde{D}_T \in \mathcal{B}} \mathbb{E}_{(\mathcal{H}, x) \sim P_{\mathcal{F}} \otimes p_{\text{data}}} \left[\mathcal{L} \left(h_{\mathcal{H}}^* \left(\tilde{D}_T \right); x \right) \right], \end{aligned}$$

meaning that this is an upper-bound relaxation of the original worst-case, max max-min objective. However, the Bayesian setting may still be more practically useful as a defender will often have some prior on the feasible student architectures.

Remark 4.3. *This Bayesian reframing transforms the original purely adversarial game into a Bayesian Robust Stackelberg game, where the defender maximizes their expected payoff against a known distribution of attacker types.*

4.3.4 Generalization of Previous Antidistillation Methods

The main distinction between different antidistillation methods is how they instantiate the set of admissible perturbations of the dataset D and in what they consider the set of viable student architectures. In the remainder of this subsection, we show how both Antidistillation Sampling ([21]) and Defensive Output Generation ([23]) fit into the framework of (4.2).

Antidistillation Sampling

ADS proceeds in an online antidistillation fashion, where individual teacher outputs are given per user query. Their formulation is a greedy approach to the outer maximization over $\tilde{D}_T$, where they try to maximize the loss of each individual teacher output. ADS considers the following perturbation to the

²Assuming $\mathcal{L}$ is non-negative and measurable, since $P_{\mathcal{F}}$ and p_{data} are independent this follows by Fubini's theorem.

teacher output distribution:

$$\tilde{p}(\cdot \mid x_{1:t}; \theta_T) := \frac{1}{Z} \exp\left(\frac{1}{\tau} \log p(\cdot \mid x_{1:t}; \theta_T) + \lambda \Delta(\cdot \mid x_{1:t}; \theta_P)\right),$$

where Z is a normalization factor, τ is the sampling temperature, $p(\cdot \mid x_{1:t}; \theta_T)$ is the original teacher distribution for token $t + 1$, λ is a scaling hyperparameter and Δ is a perturbation to the output distribution determined by the gradients of a proxy model.³

Thus, the set of admissible perturbations is the following,

$$\mathcal{B}_{\text{ADS}}(D; \theta_T) = \{\tilde{x}_i : \tilde{x}_i \sim \tilde{p}(\cdot \mid x_{1:t}; \theta_T)\}_{i=1}^n. \quad (4.6)$$

Defensive Output Generation

DOGe proposes that the task of antidistillation can be framed as the following bi-level optimization problem. The defender controls the teacher’s parameters to maximize its performance while weakening any student distilled from its outputs. The optimization problem is thus:

$$\begin{aligned} \theta_T^* &= \arg \max_{\theta_T} [\text{Perf}(\theta_T) - \lambda \cdot \text{Perf}(\theta_S^*)] \\ \theta_S^* &= \arg \min_{\theta_S} \mathcal{L}(\theta_S; D_{KD}(\theta_T)), \end{aligned}$$

where $\text{Perf}(\cdot)$ is the performance of the input model, D_{KD} is the outputs of the teacher model θ_T^* , and $\lambda > 0$ is a scaling hyperparameter.

We connect this to our formulation in (4.2) by first defining

$$\mathcal{B}_{\text{DOGe}}(D; \theta_T, \Theta_T) \quad (4.7)$$

$$= \left\{ \tilde{D}_T : \tilde{D}_T \sim p(\cdot \mid \theta_T), \theta_T \in \Theta_T, \text{Perf}(\theta_T) \geq \tau \right\}. \quad (4.8)$$

If we now consider the case where $|\mathcal{F}| = 1$, (4.2) simplifies to

$$\tilde{D}_T = \arg \max_{\tilde{D}_T \in \mathcal{B}_{\text{DOGe}}} \mathbb{E}_{x \sim p_{\text{data}}} \left[-\text{Perf}(\theta_S^*(\tilde{D}_T); x) \right].$$

Since the dataset entirely depends on the parameters θ_T we can determine

³ $\Delta(\cdot \mid x_{1:t}; \theta_P) = \ell(\theta_P^+) - \ell(\theta_P)$
 $= \ell(\theta_P + \eta \nabla_{\theta_P} \log p(x_{t+1} \mid x_{1:t}; \theta_P)) - \ell(\theta_P)$

$\tilde{D}_T$ equivalently by finding

$$\theta_T^* = \arg \max_{\theta_T \in \Theta_T, \text{Perf}(\theta_T) \geq \tau} \mathbb{E}_{x \sim p_{\text{data}}} \left[-\text{Perf}(\theta_S^*(\tilde{D}_T); x) \right],$$

and taking a Lagrangian relaxation of this problem yields

$$\theta_T^* = \arg \max_{\theta_T \in \Theta_T} \text{Perf}(\theta_T) - \lambda \text{Perf}(\theta_S^*(\theta_T))$$

which is exactly the DOGe framework.

4.4 Practical method for antidistillation

The prior works, ADS and DOGe, perturb the teacher’s output distribution (within some λ ball) at every token position, analogous to ℓ_2 attacks on images ([24, 25]) in the adversarial robustness literature. However, a well-established finding from this area is that sparse, targeted perturbations can be equally effective while being harder to detect. One-pixel attacks ([26]) and the Jacobian-based Saliency Map Approach (JSMA) ([27]) demonstrate that perturbing a small number of carefully chosen input dimensions suffices to fool classifiers, often with greater success than uniform perturbation at equivalent total distortion budgets. Motivated by this analogy, we ask: *what are the theoretical properties of an antidistillation method that modifies only $k \ll L$ tokens in a reasoning trace of length L ?* To study this, we consider a simple Gaussian perturbation scheme that admits closed-form detectability bounds.

Specifically, we consider the following constraint set $\mathcal{B}$. Let $D = \{x^{(i)}\}_{i=1}^n$ be the clean dataset generated by teacher model θ_T where each output is a sequence of length L_i (i.e. $x^{(i)} = (x_1^{(i)}, \dots, x_{L_i}^{(i)})$). Let $f_{\theta_T}(x_{<t}) \in \mathbb{R}^{|\mathcal{V}|}$ denote the pre-softmax logits of the teacher model at step t over a vocabulary $\mathcal{V}$. We define the constraint set $\mathcal{B}_{\eta,k}(D; \theta_T)$ as the set of all $\tilde{D}_T = \{\tilde{x}^{(i)}\}$ such that the following conditions hold for every sequence $i \in \{1, \dots, N\}$:

1. There exists a mask of chosen token indices $M^{(i)} \subset \{1, \dots, L_i\}$ such that $|M^{(i)}| \leq k$
2. For all $t \notin M^{(i)}$, the original token is perfectly preserved (i.e. $\tilde{x}^{(i)}_t = x^{(i)}_t$).

3. For all $t \in M^{(i)}$, the perturbed token $\tilde{x}^{(i)}$ is sampled from the following perturbed distribution

$$\tilde{x}^{(i)} \sim \text{softmax}(f_{\theta_T}(x_{<t}^{(i)}) + \xi_t), \text{ where } \xi_t \sim \mathcal{N}(0, \sigma^2 I)$$

4. We have the following bound on the variance σ^2

$$\sigma^2 \leq \frac{2\eta}{k}$$

We can see that since the constraint set perturbs some set of k tokens post-generation, the performance of the teacher model can be perfectly maintained by not perturbing the answer token(s). Broadly, the set of admissible architectures could be some set of known LLM architectures. Because we are restricting to autoregressive models, if token t is poisoned in this way, then the student model will be attempting to learn noise when it predicts token t from the previous $t - 1$ tokens. As such, this will likely degrade student performance. Thus, the main concern will be the detectability of this method. In Proposition 4.4 and Corollary 4.5 we provide results on the detectability of this method as measured by the expected Kullback-Leibler divergence between the perturbed distribution used and the original teacher distribution.

Proposition 4.4. *Let $z \in \mathbb{R}^V$ be a vector of logits defining a teacher distribution $P_T = \text{softmax}(z)$. Let $\epsilon \in \mathbb{R}^V$ be a random noise vector satisfying $\mathbb{E}[\|\epsilon\|_2^2] = \sigma^2$, and define the perturbed distribution $P_P = \text{softmax}(z + \epsilon)$. Then*

$$\mathbb{E}_\epsilon[D_{\text{KL}}(P_P \parallel P_T)] \leq \frac{\sigma^2}{2}.$$

Proof. Rewriting the KL divergence between the two softmax distributions as the Bregman divergence generated by the log-sum-exp function Φ , its integral form expresses the KL as a path integral of $\epsilon^\top \nabla^2 \Phi \epsilon$, which is a variance under the intermediate softmax and hence bounded by $\|\epsilon\|_2^2$. Evaluating $\int_0^1 t dt = \frac{1}{2}$ and taking expectation with $\mathbb{E}\|\epsilon\|_2^2 = \sigma^2$ yields the bound $\sigma^2/2$. See full proof in Appendix A. $\square$

This shows that the divergence of a single next-token distribution is bounded proportionally to the variance of the perturbation noise. In the following corollary, we show that the expected divergence for some k independent to-

kens is bounded proportionally to the number of tokens and noise added to each token.

Corollary 4.5. *If $\epsilon^{(i)} \in \mathbb{R}^V$ is applied to k independent tokens with $\mathbb{E}[\|\epsilon^{(i)}\|] \leq \sigma^2$, $\forall i$ then*

$$\mathbb{E}_\epsilon[D_{\text{KL}}(P_P \parallel P_T)] \leq k \frac{\sigma^2}{2}.$$

Proof. Since the k noisy token distributions are conditionally independent given the unchanged context, the joint KL divergence factorizes into a sum of k per-token KL divergences. We can then apply the previous proposition to each term bounds the total by $k\sigma^2/2$ (proof in Appendix A). $\square$

The main takeaway is that for any dataset in $\mathcal{B}_{\eta,k}(D; \theta_T)$ such that the mask tokens are chosen independently, the detectability is less than η . As the divergence between the perturbed distribution grows linearly with k , this shows that using a method which modifies fewer tokens will be less detectable by an attacker.

CHAPTER 5

CONCLUSION

The empirical success of Knowledge Distillation (KD) has vastly outpaced its theoretical understanding, leaving practitioners to rely on heuristics for compressing high-capacity models. This thesis sought to close that gap by providing a rigorous, information-theoretic foundation for KD. By reframing the distillation process through the lens of Rate-Distortion theory, we established that minimizing KD loss is fundamentally an optimal compression procedure, yielding the minimum achievable distortion for a given capacity constraint.

Beyond establishing these limits for simple model classes, we connected this compression optimality to the student’s performance on unseen data. By integrating our rate-distortion formulation with information-theoretic generalization bounds, we demonstrated that the capacity constraint strictly regulates the student’s generalization gap. This result helps provide fundamental limits on the performance of student models.

Crucially, understanding the mechanics of knowledge transfer also illuminates how to disrupt it. By formalizing the concept of antidistillation, we introduced a theoretical framework for protecting the intellectual property embedded within Large Language Models. Through our game-theoretic formulation, one is able to clearly see the bi-level robust optimization problem faced by the defender of distillation attacks and the issue that the defender faces of not knowing what model the student may be using to train. Furthermore, we provide a practical method for antidistillation with detection guarantees.

The framework developed in this thesis opens several promising avenues for future research. While our achievability results apply to simple model classes, extending these rate-distortion bounds to modern architectures remains an open challenge. Furthermore, analyzing the geometry of the teacher and student interaction through the lens of optimal transport and gradient

flows may yield new insights into the dynamics of how knowledge distillation functions. Ultimately, mapping the fundamental limits of both distillation and antidistillation equips the field with the theoretical tools necessary to navigate the increasingly complex landscape of model compression, deployment, and security.

APPENDIX A

PROOFS

Proof of Theorem 3.2. Let θ_T and θ_S following a joint probability distribution $P_{\theta_S, \theta_T} = P_{\theta_S|\theta_T} P_{\theta_T}$. Given a dataset with inputs drawn from a marginal distribution $X \sim P_X$ over the input space $\mathcal{X}$, let $P_\theta(Y | X)$ denote the predictive distribution of a model parameterized by θ over the label space $\mathcal{Y}$. We define the distortion function $d : \mathcal{S} \times \mathcal{T} \rightarrow \mathbb{R}^+$ as the expected Kullback-Leibler (KL) divergence between the teacher's and the student's predictive distributions:

$$d(\theta_S, \theta_T) = \mathbb{E}_{X \sim P_X} [D_{\text{KL}}(P_{\theta_T}(Y | X) \parallel P_{\theta_S}(Y | X))]. \quad (\text{A.1})$$

The information-theoretic objective is to find the optimal conditional distribution $P_{\theta_S|\theta_T}$ that minimizes the expected distortion subject to an upper bound R on the mutual information $I(\theta_T; \theta_S)$:

$$\min_{P_{\theta_S|\theta_T}} \mathbb{E}_{P_{\theta_S, \theta_T}} [d(\theta_S, \theta_T)] \quad \text{s.t.} \quad I(\theta_T; \theta_S) \leq R \quad (\text{A.2})$$

Firstly, take a Lagrangian Relaxation and introduce a Lagrange multiplier $\beta > 0$ to form the unconstrained Lagrangian functional $\mathcal{L}(P_{\theta_S|\theta_T})$. We then have that minimizing the original constrained objective is equivalent to minimizing:

$$\min_{P_{\theta_S|\theta_T}} \mathcal{L}(P_{\theta_S|\theta_T}) = \mathbb{E}_{P_{\theta_S, \theta_T}} [d(\theta_S, \theta_T)] + \beta I(\theta_T; \theta_S) \quad (\text{A.3})$$

In classical KD, the teacher is not a stochastic model but a pre-trained, frozen network with a fixed parameter vector $\theta_T^* \in \mathcal{T}$. We enforce this by setting the marginal distribution of the teacher P_{θ_T} to a Dirac measure centered at θ_T^* :

$$P_{\theta_T}(\theta) = \delta(\theta - \theta_T^*)$$

Because θ_T is now deterministic, the mutual information between the

teacher and the student is identically zero: $I(\theta_T^*; \theta_S) = 0$. Consequently, the constraint $I \leq R$ is trivially satisfied for any valid rate $R \geq 0$, and the regularization term $\beta I(\theta_T; \theta_S)$ vanishes from the Lagrangian. Furthermore, integrating over the joint distribution $\mathbb{E}_{P_{\theta_S, \theta_T}}$ collapses into an integration strictly over the student's conditional distribution given the fixed teacher:

$$\min_{P_{\theta_S|\theta_T^*}} \mathbb{E}_{\theta_S \sim P_{\theta_S|\theta_T^*}} [d(\theta_S, \theta_T^*)] \quad (\text{A.4})$$

Classical KD does not seek a posterior distribution over student weights; it seeks a maximum likelihood point estimate. We thus restrict our optimization space from the set of all valid probability measures $P_{\theta_S|\theta_T^*}$ to the subset of Dirac measures parameterized by a point $\hat{\theta}_S \in \mathcal{S}$:

$$P_{\theta_S|\theta_T^*}(\theta) = \delta(\theta - \hat{\theta}_S)$$

Substituting this Dirac measure into the expectation evaluates the integrand exactly at the point $\hat{\theta}_S$. The optimization variable shifts from a probability measure to the real-valued parameter vector $\hat{\theta}_S$:

$$\min_{\hat{\theta}_S \in \mathcal{S}} d(\hat{\theta}_S, \theta_T^*). \quad (\text{A.5})$$

Recalling equation A.1, we finally arrive at

$$\min_{\hat{\theta}_S \in \mathcal{S}} \mathbb{E}_{X \sim P_X} [D_{\text{KL}}(P_{\theta_T}(Y | X) \parallel P_{\hat{\theta}_S}(Y | X))], \quad (\text{A.6})$$

which is the original KD loss objective. $\square$

Proof of Theorem 3.4. We begin by first reducing the problem to the standard vector Gaussian R-D problem. Since x is independent of (θ_T, θ_S) , the expected distortion simplifies immediately to

$$\mathbb{E}[d(\theta_T, \theta_S)] = \mathbb{E}_{\theta_T, \theta_S} [\|\theta_T - \theta_S\|_{\Sigma_x}^2].$$

Because $\Sigma_x \succ 0$, we may introduce the invertible whitening map $\tilde{\theta} = \Sigma_x^{1/2} \theta$. Under this map the distortion becomes $\|\tilde{\theta}_T - \tilde{\theta}_S\|_2^2$, the teacher prior becomes $\tilde{\theta}_T \sim \mathcal{N}(0, \tilde{\Sigma}_T)$ with $\tilde{\Sigma}_T = \Sigma_x^{1/2} \Sigma_T \Sigma_x^{1/2}$, and mutual information is preserved by invertibility. The problem is therefore equivalent to the standard vec-

tor Gaussian rate-distortion problem for the source $\tilde{\theta}_T$ under squared-error distortion.

Now, let $\tilde{\Sigma}_T = V\Lambda V^\top$ be the eigendecomposition, with $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_d)$. Rotating to the eigenbasis via $z = V^\top \tilde{\theta}$ preserves both the Euclidean distortion and mutual information, and yields independent scalar sources $z_{T,i} \sim \mathcal{N}(0, \lambda_i)$.

Following standard procedure, we now decompose into parallel scalar problems. For independent sources $z_{T,i}$ with separable distortion, the inequality

$$I(z_T; z_S) \geq \sum_{i=1}^d I(z_{T,i}; z_{S,i})$$

holds by the chain rule and the data-processing inequality, with equality when the channel factorizes. Hence the optimal channel decomposes into d independent scalar Gaussian channels, and we allocate rates $R_i \geq 0$ with $\sum_i R_i \leq R$ to minimize $\sum_i D_i$.

Finally, we conclude by applying the classical reverse water-filling method. The scalar Gaussian rate-distortion function gives $D_i = \lambda_i e^{-2R_i}$ for the active components, with the zero-rate bound $D_i \leq \lambda_i$. Minimizing $\sum_i D_i$ subject to $\sum_i R_i \leq R$ via a Lagrange multiplier $\mu > 0$ yields the classical *reverse water-filling* solution $D_i = \min(\mu, \lambda_i)$, where μ is chosen to satisfy the rate constraint. Summing over all dimensions gives the stated parametric equations

$$R(\mu) = \sum_{i=1}^d \frac{1}{2} \max\left(0, \ln \frac{\lambda_i}{\mu}\right),$$

$$D(\mu) = \sum_{i=1}^d \min(\mu, \lambda_i). \quad \square$$

Proof of Lemma 3.5. It suffices to verify that each active component ($\lambda_i > \mu$) achieves exactly $D_i = \mu$ and $R_i = \frac{1}{2} \ln(\lambda_i/\mu)$, since the inactive components trivially achieve $D_i = \lambda_i$ at zero rate by setting $\hat{z}_i = 0$.

Write the channel as $\hat{z}_i = \alpha_i z_{T,i} + w_i$ with $\alpha_i = 1 - \mu/\lambda_i$ and $\sigma_{w_i}^2 = \mu(1 - \mu/\lambda_i)$.

Distortion. By independence of w_i and $z_{T,i}$:

$$\mathbb{E}[(z_{T,i} - \hat{z}_i)^2] = (1 - \alpha_i)^2 \lambda_i + \sigma_{w_i}^2 = \frac{\mu^2}{\lambda_i} + \mu \left(1 - \frac{\mu}{\lambda_i}\right) = \mu. \quad (\text{A.7})$$

Rate. We first compute $\text{Var}(\hat{z}_i)$:

$$\text{Var}(\hat{z}_i) = \alpha_i^2 \lambda_i + \sigma_{w_i}^2 = \left(1 - \frac{\mu}{\lambda_i}\right) \left[\lambda_i \left(1 - \frac{\mu}{\lambda_i}\right) + \mu\right] = \lambda_i \left(1 - \frac{\mu}{\lambda_i}\right). \quad (\text{A.8})$$

Since the channel is Gaussian, the mutual information is:

$$I(z_{T,i}; \hat{z}_i) = \frac{1}{2} \ln \frac{\text{Var}(\hat{z}_i)}{\sigma_{w_i}^2} = \frac{1}{2} \ln \frac{\lambda_i (1 - \mu/\lambda_i)}{\mu (1 - \mu/\lambda_i)} = \frac{1}{2} \ln \frac{\lambda_i}{\mu}. \quad (\text{A.9})$$

Both the distortion and rate targets are met exactly, so the channel is feasible and optimal.

Uniqueness. The Gaussian rate-distortion bound for a scalar Gaussian source under squared-error distortion is achieved uniquely by the Gaussian test channel [28]. Since the problem decomposes into d independent scalar Gaussian problems (as shown in the proof of Theorem 1), the optimal channel is unique in each coordinate, and therefore unique overall. $\square$

Proof of Theorem 3.10. **1. Quadratic Approximation of the Distortion**

We begin by expanding the KL divergence around θ . Because $D_{KL}(P \parallel P) = 0$ and its first derivative with respect to the second argument evaluated at the first is zero, the leading term is the second-order term (the Hessian). For exponential family distributions, the Hessian of the KL divergence is the Fisher Information Matrix H . Taking the expectation over the data x , the expected distortion becomes a Mahalanobis distance:

$$d(\theta, \theta') \approx \frac{1}{2} (\theta - \theta')^T H (\theta - \theta')$$

2. Whitening the Prior (Decoupling)

We define a coordinate transformation to simultaneously diagonalize the prior covariance and the FIM. Let z be an isotropic Gaussian random vector such that $\theta = \Sigma_\theta^{1/2} z$. It follows that $z \sim \mathcal{N}(0, I)$. Similarly, we transform the

student parameters: $\theta' = \Sigma_\theta^{1/2} z'$. Substituting these into the distortion yields:

$$d(z, z') = \frac{1}{2}(z - z')^T \left(\Sigma_\theta^{1/2} H \Sigma_\theta^{1/2} \right) (z - z')$$

Let $M = \Sigma_\theta^{1/2} H \Sigma_\theta^{1/2}$. Since M is symmetric and positive-definite, it admits an orthogonal eigendecomposition $M = V \Lambda V^T$, where V is an orthogonal matrix ($V V^T = I$) and $\Lambda = \text{diag}(\gamma_1, \dots, \gamma_k)$ contains the eigenvalues.

3. Projection into Eigenspace

We apply the orthogonal transformation V^T to define new variables: $w = V^T z$ and $w' = V^T z'$. Because V is an orthogonal rotation, the prior on w remains standard normal: $w \sim \mathcal{N}(0, I)$. The distortion now completely decouples into a sum of k independent scalar channels:

$$d(w, w') = \frac{1}{2}(w - w')^T \Lambda (w - w') = \sum_{i=1}^k \frac{1}{2} \gamma_i (w_i - w'_i)^2$$

Taking the expectation over the joint distribution $p(w, w')$, the total expected distortion D is:

$$D = \sum_{i=1}^k \frac{1}{2} \gamma_i D_i$$

where $D_i = \mathbb{E}[(w_i - w'_i)^2]$ is the mean squared error in the i -th channel.

Because the components of w are independent, mutual information satisfies $I(w; w') \geq \sum_{i=1}^k I(w_i; w'_i)$. To minimize distortion for a given rate, the channels must be conditionally independent, allowing us to write the total rate as:

$$R = \sum_{i=1}^k R_i = \sum_{i=1}^k I(w_i; w'_i)$$

4. Rate-Distortion Optimization

For a scalar Gaussian source $w_i \sim \mathcal{N}(0, 1)$, the rate-distortion function is $R_i(D_i) = \frac{1}{2} \max(0, \log \frac{1}{D_i})$ subject to $D_i \leq \mathbb{E}[w_i^2] = 1$. We formulate the unconstrained Lagrangian with multiplier $\mu > 0$ to minimize the joint distortion and rate:

$$\mathcal{L} = \sum_{i=1}^k \left(\frac{1}{2} \gamma_i D_i + \mu R_i \right) = \sum_{i=1}^k \left(\frac{1}{2} \gamma_i D_i - \frac{\mu}{2} \log D_i \right)$$

Taking the derivative with respect to D_i for active channels ($D_i < 1$) and

setting it to zero:

$$\frac{\partial \mathcal{L}}{\partial D_i} = \frac{1}{2}\gamma_i - \frac{\mu}{2D_i} = 0 \implies D_i = \frac{\mu}{\gamma_i}$$

Enforcing the strict upper bound $D_i \leq 1$ (the student cannot have more error than simply predicting the prior mean zero), we obtain the optimal allocation:

$$D_i^* = \min\left(1, \frac{\mu}{\gamma_i}\right)$$

5. Final Parametric Expressions

Substituting the optimal channel distortions D_i^* back into the equations for R and D :

$$R(\mu) = \sum_{i=1}^k \frac{1}{2} \max\left(0, \log \frac{1}{D_i^*}\right) = \frac{1}{2} \sum_{i=1}^k \max\left(0, \log \frac{\gamma_i}{\mu}\right)$$

$$D(\mu) = \sum_{i=1}^k \frac{1}{2} \gamma_i D_i^* = \frac{1}{2} \sum_{i=1}^k \gamma_i \min\left(1, \frac{\mu}{\gamma_i}\right) = \frac{1}{2} \sum_{i=1}^k \min(\gamma_i, \mu)$$

This concludes the proof. $\square$

Proof of Proposition 4.4. We begin by defining the log-sum-exp function

$$\Phi(z) = \log \sum_{w \in \mathcal{V}} \exp(z_w).$$

Φ has two important properties:

$$\nabla \Phi(z) = P_T, \tag{A.10}$$

$$\nabla^2 \Phi(z) = \text{diag}(P_T) - P_T P_T^\top. \tag{A.11}$$

Because $\log P_T(v) = z_v - \Phi(z)$ and $\log P_P(v) = z_v + \epsilon_v - \Phi(z + \epsilon)$, the KL divergence expands as

$$\begin{aligned} D_{\text{KL}}(P_P \parallel P_T) &= \sum_v P_P(v) [\log P_P(v) - \log P_T(v)] \\ &= \sum_v P_P(v) [\epsilon_v - \Phi(z + \epsilon) + \Phi(z)] \\ &= \Phi(z) - \Phi(z + \epsilon) + \langle \nabla \Phi(z + \epsilon), \epsilon \rangle. \end{aligned}$$

Rewriting $\epsilon = z - (z + \epsilon)$ in the inner product recovers the definition of the Bregman divergence generated by Φ :

$$D_{\text{KL}}(P_P \parallel P_T) = \Phi(z) - \Phi(z + \epsilon) - \langle \nabla \Phi(z + \epsilon), z - (z + \epsilon) \rangle = D_{\Phi}(z, z + \epsilon).$$

We know that any Bregman divergence admits the integral representation

$$D_{\Phi}(z, z + \epsilon) = \int_0^1 t \epsilon^\top \nabla^2 \Phi(z + t\epsilon) \epsilon dt,$$

where the path is the line segment $z + t\epsilon$ for $t \in [0, 1]$. Writing $P_t = \nabla \Phi(z + t\epsilon)$ for the intermediate softmax distribution, this gives

$$D_{\text{KL}}(P_P \parallel P_T) = \int_0^1 t \epsilon^\top (\text{diag}(P_t) - P_t P_t^\top) \epsilon dt.$$

The quadratic form equals the variance of the components of ϵ under P_t :

$$\epsilon^\top (\text{diag}(P_t) - P_t P_t^\top) \epsilon = \sum_v P_t(v) \epsilon_v^2 - \left(\sum_v P_t(v) \epsilon_v \right)^2 = \text{Var}_{P_t}(\epsilon).$$

Now we also have,

$$\text{Var}_{P_t}(\epsilon) \leq \sum_v P_t(v) \epsilon_v^2 \leq \sum_v \epsilon_v^2 = \|\epsilon\|_2^2.$$

Hence, uniformly for all $t \in [0, 1]$,

$$\epsilon^\top \nabla^2 \Phi(z + t\epsilon) \epsilon \leq \|\epsilon\|_2^2.$$

Substituting in the bound we get,

$$D_{\text{KL}}(P_P \parallel P_T) \leq \int_0^1 t \|\epsilon\|_2^2 dt.$$

Since $\|\epsilon\|_2^2$ no longer depends on t or on the intermediate distributions P_t , we may pass the expectation through the integral:

$$\begin{aligned} \mathbb{E}_\epsilon[D_{\text{KL}}(P_P \parallel P_T)] &\leq \mathbb{E}_\epsilon \left[\int_0^1 t \|\epsilon\|_2^2 dt \right] = \int_0^1 t \mathbb{E}_\epsilon[\|\epsilon\|_2^2] dt \\ &= \sigma^2 \int_0^1 t dt = \sigma^2 \cdot \frac{1}{2} = \frac{\sigma^2}{2}. \end{aligned} \quad \square$$

Proof of Corollary 4.5. Observe that since the k token distributions that are independently perturbed by noise are themselves independently conditioned on the original unchanged context (because the noise is added post-hoc, not during generation), we have that

$$P_P(y_M|x, y_{\setminus M}) = \prod_{i=1}^k P_P(y_{m_i}|x, y_{\setminus M}).$$

Similarly,

$$P_T(y_M|x, y_{\setminus M}) = \prod_{i=1}^k P_T(y_{m_i}|x, y_{\setminus M})$$

Thus, we have that

$$D_{\text{KL}}(P_P(y_M) \parallel P_T(y_M)) = \sum_{i=1}^k D_{\text{KL}}(P_P(y_{m_i}) \parallel P_T(y_{m_i})).$$

Taking expectations, we have that

$$\begin{aligned} \mathbb{E} \left[\sum_{i=1}^k (P_P(y_M) \parallel P_T(y_M)) \right] &= \sum_{i=1}^k \mathbb{E} [D_{\text{KL}}(P_P(y_{m_i}) \parallel P_T(y_{m_i}))] \\ &\leq \sum_{i=1}^k \frac{\sigma^2}{2} = k \frac{\sigma^2}{2} \end{aligned}$$

□

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